

On semiorthogonal decompositions

for twisted flag varieties

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k -field, all varieties are smooth.

Thm (Beilinson '78) $D^b(\mathbb{P}^n) = \langle D^b(k) \otimes \mathcal{O}(-n), D^b(k) \otimes \mathcal{O}(-n+1), \dots, D^b(k) \otimes \mathcal{O} \rangle$ - is a semiorthogonal decomposition

Def. $\mathcal{E} = \langle \mathcal{E}_1, \dots, \mathcal{E}_n \rangle$ - s.o.d. if 1) $\text{Hom}(\mathcal{E}_i, \mathcal{E}_j) = 0 \ \forall \mathcal{E}_i \in \mathcal{E}_i, \mathcal{E}_j \in \mathcal{E}_j, i > j$
 triang. cat. full triang. subcats 2) $\forall M \in \mathcal{E} \exists$ filtration $0 \rightarrow M_{n-1} \xrightarrow{d_{n-1}} M_{n-2} \xrightarrow{d_{n-2}} \dots \xrightarrow{d_2} M_1 \xrightarrow{d_1} M$ st. $\text{cone}(d_i) \in \mathcal{E}_i$

Def $M_1, \dots, M_n \in \mathcal{E}$ - full exceptional collection is

- 1) $\text{Hom}(M_i, M_j[m]) = \begin{cases} k, m=0 \\ 0, m \neq 0 \end{cases}$
- 2) $\text{Hom}(M_i, M_j[m]) = 0, i > j$
- 3) $\{M_1, \dots, M_n\}$ generate $\mathcal{E}$ as a triang. cat.

Claim: X - smooth proj, $M_1, \dots, M_n \in D^b(X)$ - full ex. collection $\Rightarrow D^b(X) = \langle D^b(k) \otimes M_1, \dots, D^b(k) \otimes M_n \rangle$ - s.o.d.

Kapranov '84 & '88 - failure for Grassmannians and quadrics.

Conjecture: G - split semisimple, $P \leq G$ - parabolic $\Rightarrow D^b(G/P)$ admits a full ex. coll.

Orlov, Kuznetsov, Kuznetsov, Forrester, ... many examples.

Samokhin - van der Kallen '24 - conjecture holds.

Question: • what happens in families? i.e. for $E/P \rightarrow S$, where $E \rightarrow S$ is a G -torsor.
 • in particular, for $S = \text{Spec } k$: E/P - twisted form of G/P ($\cong G/P$ over E)

Thm (Bernardara '05) $\mathcal{A}/S$ - sheaf of Azumaya algebras $\rightarrow \text{SB}(\mathcal{A}) \rightarrow S$ - Severi-Brauer variety (a form of $\mathbb{P}^n_S$)
 $\Rightarrow D^b(\text{SB}(\mathcal{A})) = \langle D^b(S), D^b(S, \mathcal{A}), \dots, D^b(S, \mathcal{A}^{\otimes n}) \rangle$
 $D^b(\mathcal{A}\text{-mod}) \cong$ twisted sheaves

Baek '12 - similar for Grassmannians

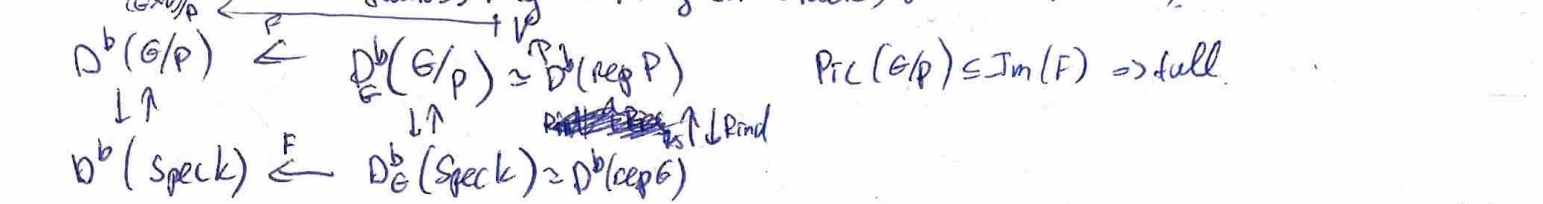
Thm (Panin '94) G - quasi-split group, E - G -torsor, $P \leq G$ - parabolic has a Borel subgroup $\Rightarrow K_*(E/P) \cong \bigoplus_{i=1}^n K_*(A_i), A_i/k_i$ - i.e. a

Rem. X - projective variety, homogeneous under a semisimple group $\Rightarrow X \cong E/P$ as above. K_i/k_i - lin. sep. field extensions $\sum_{i=1}^n [k_i:k] = [w(E):w(P)]$

Thm (Samokhin - vd Kallen '24)

G - split simply connected semisimple, $P \leq G$ parabolic $\Rightarrow \exists X_i \in D^b(\text{rep } P)$ s.t.
 $D^b(\text{rep } P) = \langle D^b(\text{rep } G) \otimes X_1, \dots, D^b(\text{rep } G) \otimes X_n \rangle$ - s.o.d.
 $D^b(BP) = \langle D^b(BG) \otimes X_1, \dots, D^b(BG) \otimes X_n \rangle$
 $G/P \xrightarrow{\text{BP}} BG$ - universal G/P bundle

These X_i form a relative ex. collation ($\text{Rind}_B^C \text{Hom}(X_i, X_j) = \begin{bmatrix} 0 & i \rightarrow j \\ k & i=j \end{bmatrix}$) which is full.
 (D^b(rep G) ⊗ X_i gen. D^b(rep P))



Thm (A. Samokhin) $E \rightarrow S$ - G -torsor $\Rightarrow D^b(E/P) = \langle D^b(S) \otimes X_1^E, \dots, D^b(S) \otimes X_n^E \rangle$ - s.o.d.
Pf. $D^b(E/P) \simeq D^b_G(E \times G/P) \xleftarrow{X_i^E} D^b_G(G/P)$ • X_i^E - relative ex. collection by flat base change
 $\uparrow \downarrow$ $\uparrow \downarrow$ $\uparrow \downarrow$
 $D^b(S) \simeq D^b_G(E) \xleftarrow{X_i^E} D^b_G(\text{Spec } k)$ • full because $\omega_{E/P/S}$ is relatively ample and in the image.

Azumaya algebras appear when G is not s.c.

Thm (A. Samokhin) G -split semisimple, $P \leq G$ -parabolic. $\Rightarrow$
 $\Rightarrow D^b(\text{rep } P) = \langle D^b(A_1\text{-mod}), \dots, D^b(A_n\text{-mod}) \rangle$
 for some $A_i \simeq M_{n_i}(k)$ with G -action $D^b(BG, A_i)$ - Azumaya alg. over BG

Pf. $\tilde{G} \xrightarrow{P} G$ - s.c. cover, $Z := \ker P$ - diagonalizable group, central in $\tilde{G}$
 $\tilde{P} \rightarrow P \Rightarrow \text{rep } \tilde{G} = \bigoplus_{\lambda \in X^*(Z)} (\text{rep } \tilde{G})_\lambda$ $\Rightarrow \text{rep } Z$ is semisimple
 $D^b(\text{rep } \tilde{G}) = \bigoplus_{\lambda \in X^*(Z)} D^b(\text{rep } \tilde{G})_\lambda$ $\leftarrow \text{repr. of } \tilde{G} \text{ where } Z \text{ acts via } \lambda$
 $D^b(\text{rep } \tilde{G}) \simeq D^b(\text{rep } \tilde{G})_0$
 & similarly for $\tilde{P}$ $D^b(\text{rep } \tilde{G})_\lambda$.

$D^b(G/P) = \langle D^b(G) \otimes X_1, \dots, D^b(G) \otimes X_n \rangle$ $X_i \in D^b(\text{rep } \tilde{P})_{\lambda_i}$ in fact, these are Steinberg weights (give a basis of $R(\tilde{P})$ over $R(G)$)
 $\Rightarrow D^b(\text{rep } P) = \langle D^b(\tilde{G})_{-\lambda_i} \otimes X_1, \dots, D^b(\tilde{G})_{-\lambda_n} \otimes X_n \rangle$
 pick $0 \neq \nabla_{\lambda_i} \in (\text{rep } \tilde{G})_{\lambda_i} \rightsquigarrow A_i := \text{End}_{AG(k)}(\nabla_{\lambda_i})$, $Y_i := X_i \otimes \nabla_{\lambda_i}^* \in D^b(\text{rep } \tilde{P})_0 \simeq D^b(\text{rep } P)$
 $\Rightarrow D^b(A_i\text{-mod}) \rightarrow D^b(\text{rep } P)$ - f.g., give s.o.d.
 $M \mapsto Y_i \otimes M$ $\text{rem. (Elagin 03)} D^b(\tilde{G})_{\lambda_i} \simeq D^b(A_i\text{-mod})$

In fact, Y_i form a relative Azumaya-exceptional collection: $\text{Rind Hom}(Y_i, Y_j) = \begin{bmatrix} 0 & i \rightarrow j \\ A_i & i=j \end{bmatrix}$
Cor. G -split semisimple, $P \leq G$ -parabolic, $E \rightarrow S$ - G -torsor
 $\Rightarrow D^b(E/P) = \langle D^b(S, A_1^E), \dots, D^b(S, A_n^E) \rangle$ - s.o.d.
 "Azumaya alg. $\hat{A}_i$ "

Similarly for quasi-split groups (in progress)